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Digital certificate management

Abstract

According to one embodiment, a mobile communication apparatus includes a communication module and a main module. The communication module is capable of storing only one of an MQTTS certificate and an HTTPS certificate. The main module is capable of storing both the MQTTS certificate and the HTTPS certificate. The main module provides the MQTTS certificate to the communication module if the communication module starts MOTTS communication. The main module provides the HTTPS certificate to the communication module if the communication module starts HTTPS communication in a case where the communication module performs the HTTPS communication during a keep-alive interval of the MQTTS communication and provides the MQTTS certificate to the communication module if the communication module stops the HTTPS communication.

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Background/Summary

FIELD
(1) Embodiments described herein relate generally to a mobile communication apparatus, a network system, and a communication method.
BACKGROUND
(2) In the related art, in case of using certificates for various protocols, there is a certificate for each protocol. For example, in the case of a system that manages devices as mobile device management, the device is managed with MOTTS, and HTTPS is used for file communication. A certificate is generally used for each for ensuring security. However, some communication apparatuses have only one certificate storage area. Since such communication apparatuses use the same certificate for a plurality of protocols, the security level is lowered.

Description

DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating a configuration of a network system including a mobile communication apparatus according to an embodiment; and
(2) FIG. **2** is a flowchart illustrating an operation of the network system illustrated in FIG. **1**.
DETAILED DESCRIPTION
(3) In general, according to one embodiment, a mobile communication apparatus includes a

communication module and a main module. The communication module is capable of storing only one of an MOTTS certificate and an HTTPS certificate. The main module is capable of storing both the MOTTS certificate and the HTTPS certificate. The main module provides the MQTTS certificate to the communication module if communication the module starts MOTTS communication, and the communication module uses the received MQTTS certificate to start the MQTTS communication. The main module provides the HTTPS certificate to the communication module if the communication module starts HTTPS communication in a case where the communication module performs the HTTPS communication during a keep-alive interval of the MQTTS communication, the communication module uses the received HTTPS certificate to start the HTTPS communication, the main module provides the MQTTS certificate to the communication module if the communication module stops the HTTPS communication, and the communication module uses the received MQTTS certificate to transmit a PINGREQ command and to continue the MOTTS communication.
(4) Hereinafter, the embodiment is described below with reference to FIG. **1**. FIG. **1** is a diagram illustrating a configuration of a network system including the mobile communication apparatus according to the embodiment. In the embodiment, the mobile communication apparatus is a mobile printer **10**.
(5) Network System
(6) The network system includes the mobile printer **10**, an access point **50**, an MQTT broker **60**, mobile device management **70**, an HTTPS server **80**, and a dispensing terminal **90**.
(7) The access point **50** is a connection destination for wireless communication by the mobile printer **10**.
(8) The MQTT broker **60** wirelessly communicates with the mobile printer **10** via the access point **50**, establishes the MQTTS communication with the mobile device management **70**, and is a relay server for establishing the HTTPS communication with the HTTPS server **80**.
(9) The mobile device management **70** manages the mobile printer **10** via the MQTT broker **60** and the access point **50** by the MQTTS communication. For example, the mobile device management **70** manages parameters, voltages, and states (idle or active) of the mobile printer **10** and the like.
(10) The HTTPS server **80** stores communication data required by the mobile printer **10**, for example, update data for the firmware of the mobile printer **10**. The communication data is downloaded to the mobile printer **10** via the MQTT broker **60** and the access point **50** by the HTTPS communication.
(11) The dispensing terminal **90** is a terminal that generates print data of the mobile printer **10** and transmits the print data, a print command, and the like. The transmission data of the dispensing terminal **90** is transmitted to the mobile printer **10** via the access point **50** by SOCKET communication.
(12) Mobile Printer
(13) The mobile printer **10** includes a main module **20** and a communication module **30**. The main module **20** and the communication module **30** can communicate with each other.
(14) The main module **20** includes a CPU **21**, a flash ROM **22**, an SDRAM **23**, a printer **24**, and an operation panel **25**.
(15) The CPU **21** is a processor and controls the flash ROM **22**, the SDRAM **23**, the printer **24**, and the operation panel **25**.
(16) The flash ROM **22** is a non-volatile memory that non-temporarily stores an MOTTS certificate **41** and an HTTPS certificate **42** both. The flash ROM **22** also stores firmware, a control program, and the like.
(17) The SDRAM **23** is a volatile memory that temporarily stores the firmware, the control program, and the like executed by the CPU **21**.
(18) The printer **24** is a device that prints the print data received from the dispensing terminal **90** under the control of the CPU **21** if the mobile printer **10** receives the print command from the

dispensing terminal **90**. For example, the printer **24** is a thermal printer.
(19) The operation panel **25** is a device that receives a command from a user and displays information to a user for promotion.
(20) The CPU **21** reads the firmware stored in the flash ROM **22** to the SDRAM **23** and executes the firmware for controlling the entire main module **20**.
(21) The communication module **30** includes a communication circuit **31**, a CPU **32**, a flash ROM **33**, and an SDRAM **34**.
(22) The communication circuit **31** is a circuit for communication with the outside. Specifically, the communication circuit **31** performs the SOCKET communication with the dispensing terminal **90** via the access point **50**. In addition, the communication circuit **31** performs the MQTTS communication with the mobile device management **70** and performs the HTTPS communication with the HTTPS server **80**, via the access point **50** and the MQTT broker **60**.
(23) The CPU **32** is a processor and controls the communication circuit **31**, the flash ROM **33**, and the SDRAM **34**.
(24) The flash ROM **33** is a non-volatile memory that can non-temporarily store only one certificate **43** in order to reduce the size of the communication module **30**. The certificate **43** is any one of the MQTTS certificate **41** and the HTTPS certificate **42**. The flash ROM **33** also stores firmware.
(25) The SDRAM **34** is a volatile memory that temporarily stores the firmware executed by the CPU **32**.
(26) The CPU **32** reads the firmware stored in the flash ROM **33** to the SDRAM **34** and executes the firmware for controlling the entire communication module **30**.
(27) Operation
(28) Hereinafter, with reference to FIG. **2**, an operation of the network system illustrated in FIG. **1** is described. FIG. **2** is a flowchart illustrating the operation of the network system illustrated in FIG. **1**. In the following description, an operation of the mobile printer **10** is performed by the main module **20**. Specifically, the CPU **21** of the main module **20** reads the control program stored in the flash ROM **22** to the SDRAM **23** and executes the control program to perform the operation of the mobile printer **10**.
(29) In Act **1**, at the start of the mobile printer **10**, the main module **20** provides the MOTTS certificate **41** to the communication module **30**. The communication module **30** stores the received MOTTS certificate **41** in the flash ROM **33**. The communication module **30** uses the MOTTS certificate **41** to transmit a CONNECT command and to be connected to the MQTT broker **60** via the access point **50**. Similarly, the MQTT broker **60** establishes the MOTTS communication with the mobile device management **70**. As a result, the communication module **30** starts the MOTTS communication with the mobile device management **70** via the access point **50** and the MQTT broker **60**. After the start of the MOTTS communication, the mobile device management **70** manages the mobile printer **10** by the MOTTS communication via the MQTT broker **60** and the access point **50**.
(30) The MQTT broker **60** cuts the connection to the mobile printer **10** if there is no communication from the mobile printer **10** for 1.5 times a predetermined time interval. In order to continue the connection to the MQTT broker **60**, the communication module **30** periodically transmits the PINGREQ command at a predetermined time interval. The periodical transmission of the PINGREQ command is referred to as keep-alive. Hereinafter, the predetermined time interval is referred to as the keep-alive interval, for convenience. A set value of the keep-alive interval is included in the CONNECT command as a parameter. For example, the set value of the keep-alive interval is 60 seconds.
(31) In Act **2**, during the keep-alive interval of the MOTTS communication, the main module **20** provides the HTTPS certificate **42** to the communication module **30**. The communication module **30** stores the received HTTPS certificate **42** in the flash ROM **33**. That is, the communication

module **30** replaces the MQTTS certificate **41** in the flash ROM **33** with the HTTPS certificate **42**. Thereafter, the communication module **30** uses the HTTPS certificate **42** to be connected to the MQTT broker **60** via the access point **50**. Accordingly, the MQTT broker **60** establishes the HTTPS communication with the HTTPS server **80**. That is, the communication module **30** starts the HTTPS communication with the HTTPS server **80** via the access point **50** and the MQTT broker **60**.
(32) In Act **3**, the main module **20** confirms whether there is communication data, such as update data for the firmware in the HTTPS server **80** via the communication module **30**, the access point **50**, and the MQTT broker **60**.
(33) As a result of the confirmation in Act **3**, if there is the communication data in the HTTPS server **80**, the main module **20** determines whether to increase the keep-alive interval in Act **4**. If a data amount of the communication data is large, for example, if it is expected that the download does not stop during the current keep-alive interval or several subsequent keep-alive intervals, the main module **20** determines that it is necessary to increase the keep-alive interval.
(34) If it is determined that the keep-alive interval needs to be increased, in Act **4**, the main module **20** changes the set value of the keep-alive interval to increase the set value, in Act **5**.
(35) In Act **6**, the main module **20** downloads the communication data from the HTTPS server **80** via the communication module **30**, the access point **50**, and the MQTT broker **60**.
(36) In Act **7**, the communication module **30** stops the HTTPS communication during the keep-alive interval. Subsequently, the main module **20** provides the MQTTS certificate **41** to the communication module **30**. The communication module **30** stores the received MOTTS certificate **41** in the flash ROM **33**. That is, the communication module **30** replaces the HTTPS certificate **42** in the flash ROM **33** with the MQTTS certificate **41**.
(37) Subsequently, in Act **8**, for example, at the end of the keep-alive interval, the communication module **30** uses the MQTTS certificate **41** to transmit the PINGREQ command and to continue the connection to the MQTT broker **60** via the access point **50**. The PINGREQ command includes the set value after the change of the keep-alive interval, that is, after the increase thereof, as a parameter.
(38) If it is determined that the keep-alive interval does not need to be increased, in Act **4**, the main module **20** skips the process of Act **5** and proceeds to the downloading process of Act **6**.
(39) The processes of Acts **4** and **5** are not necessarily required and may be skipped.
(40) As a result of the confirmation in Act **3**, if there is no communication data in the HTTPS server **80**, the main module **20** skips the processes of Acts **4** to **6** and proceeds to the certificate replacement process of Act **7**.
(41) Thereafter, the mobile printer **10** (the main module **20** and the communication module **30**) repeats the processes of Acts **2** to **8** until the downloading of the communication data is completed. That is, the mobile printer **10** continues the MQTTS communication by replacing the MOTTS certificate **41** and the HTTPS certificate **42** with each other during the keep-alive interval of the MQTTS communication and downloads the communication data by the HTTPS communication.
(42) After the downloading of the communication data is completed, the mobile printer **10** may skip the processes of Acts **2** to **7** and may return the set value of the keep-alive interval to an initial value.
(43) The mobile device management **70** continues the management of the mobile printer **10** until the MQTTS communication is interrupted by shutdown of the mobile printer **10** or the like.
Effect
(44) In the mobile printer **10** according to the embodiment, the communication module **30** can store only one certificate **43** but can perform the MQTTS communication and the HTTPS communication without lowering the security level by providing one of the MQTTS certificate **41** and the HTTPS certificate **42** appropriate for the required communication to the communication module **30** by the main module **20**.

(45) In addition, the number of times of starting and stopping the HTTPS communication is reduced by increasing the keep-alive interval of the MQTTS communication, so that the reduction of the entire time required for downloading the communication data can be attempted.
(46) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel apparatus, system and method described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A mobile communication apparatus, comprising: a communication module configured to store only one of an MQTTS certificate and an HTTPS certificate, use the MQTTS certificate to perform MQTTS communication, and use the HTTPS certificate to perform HTTPS communication; and a main module configured to store both the MQTTS certificate and the HTTPS certificate, provide the MQTTS certificate to the communication module if the communication module starts the MQTTS communication, to enable the communication module to start the MQTTS communication, provide the HTTPS certificate to the communication module if the communication module starts the HTTPS communication in a case where the communication module performs the HTTPS communication during a keep-alive interval of the MQTTS communication, to enable the communication module to start the HTTPS communication, and provide the MQTTS certificate to the communication module if the communication module stops the HTTPS communication, to enable the communication module to transmit a PINGREQ command and continue the MQTTS communication, wherein the communication module repeats the HTTPS communication throughout a plurality of the keep-alive intervals if a communication amount of the HTTPS communication is large, wherein the communication module increases a set value of the keep-alive interval and increases a period of the HTTPS communication, wherein the main module changes the set value of the keep-alive interval to increase the set value, and wherein the main module returns the set value of the keep-alive interval to an initial value in response to a download of communication data being determined to be complete.
2. The mobile communication apparatus according to claim 1, wherein the main module stores both the MQTTS certificate and the HTTPS certificate in a flash memory.
3. The mobile communication apparatus according to claim 1, wherein the mobile communication apparatus is a mobile printer.
4. A network system, comprising: a mobile communication apparatus that includes a communication module configured to store only one of an MQTTS certificate and an HTTPS certificate, use the MQTTS certificate to perform MQTTS communication, and use the HTTPS certificate to perform HTTPS communication, and a main module configured to store both the MQTTS certificate and the HTTPS certificate, provide the MQTTS certificate to the communication module if the communication module starts the MQTTS communication, to enable the communication module to start the MQTTS communication, provide the HTTPS certificate to the communication module if the communication module starts the HTTPS communication in a case where the communication module performs the HTTPS communication during a keep-alive interval of the MQTTS communication, to enable the communication module to start the HTTPS communication, and provide the MQTTS certificate to the communication module if the communication module stops the HTTPS communication, to enable the communication module to transmit a PINGREQ command and continue the MQTTS communication; an access point comprising a connection destination for wireless communication by the mobile communication

apparatus; an MQTT broker that wirelessly communicates with the mobile communication apparatus via the access point; mobile device management that manages the mobile communication apparatus via the MQTT broker and the access point by the MQTTS communication; and an HTTPS server that stores communication data to be downloaded to the mobile communication apparatus via the MQTT broker and the access point by the HTTPS communication, wherein the communication module repeats the HTTPS communication throughout a plurality of the keep-alive intervals if a communication amount of the HTTPS communication is large, wherein the communication module increases a set value of the keep-alive interval and increases a period of the HTTPS communication, wherein the main module changes the set value of the keep-alive interval to increase the set value, and wherein the main module returns the set value of the keep-alive interval to an initial value in response to a download of communication data being determined to be complete.

5. The network system according to claim 4, wherein the main module stores both the MQTTS certificate and the HTTPS certificate in a flash memory.

6. The network system according to claim 4, wherein the mobile communication apparatus is a mobile printer.

7. A communication method for causing a mobile communication apparatus that includes a communication module configured to store only one of an MQTTS certificate and an HTTPS certificate and a main module configured to store both the MQTTS certificate and HTTPS certificate to perform: providing the MQTTS certificate from the main module to the communication module if MQTTS communication is started, to enable the communication module to start the MQTTS communication; providing the HTTPS certificate from the main module to the communication module if HTTPS communication is started in a case where the HTTPS communication is performed during a keep-alive interval of the MQTTS communication, to enable the communication module to start the HTTPS communication; providing the MQTTS certificate from the main module to the communication module if the HTTPS communication is stopped, to enable the communication module to transmit a PINGREQ command and continue the MQTTS communication; repeating the HTTPS communication throughout a plurality of the keep-alive intervals if a communication amount of the HTTPS communication is large; increasing a set value of the keep-alive interval and increasing a period of the HTTPS communication; changing the set value of the keep-alive interval to increase the set value; and returning the set value of the keep-alive interval to an initial value in response to a download of communication data being determined to be complete.

8. The communication method according to claim 7, further comprising: storing both the MQTTS certificate and the HTTPS certificate in a flash memory.

9. The communication method according to claim 7, wherein the mobile communication apparatus is a mobile printer.